

Data Management and File Structures - I

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COURSE CONTENT

- Database System: Concepts and Architecture (ch1-2)
- Data Modeling Using the Entity-Relationship (ER) Model (ch3)
- Enhanced Entity-Relationship (EER) Modeling (ch4)
- The Relational Data Model and Relational Database Constraints (ch5)
- ER- and EERR-to-Relational Mapping (ch 7)
- Functional Dependencies and Normalisation for Relational Databases (ch 10)
- Query Representation; Query Types; The Relational Algebra and Calculus (ch 6 – page 1-77)
- Disk Storage, Basic File Structures, and Hashing (ch 13)
- Indexing Structures (ch 14)

RESOURCES

- Elmasri, Ramez, and Shamkant B. Navathe. *Fundamentals of Database Systems, Global Edition*. Pearson Education Limited, 2015.
- Kandiri, John. "Principles of Database Systems." (2018).

RESOURCES

- S. B. Navathe, R. A. Elmasri, "Fundamentals of Database Systems", Addison-Wesley
- Date, "An Introduction to Database Systems", Addison-Wesley
- Hoffer, Venkataraman, Topi, Modern Database Management, Prentice Hall
- D. M. Kroenke, "Database Processing: Fundamentals, Design and Implementation", Prentice Hall

RESOURCES

- Date, Darwen, “Databases- Types and the Relational Model”, Addison-Wesley
- J. Ullman, J. Widom, “A First Course in Database Systems”
- ...

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WEEKLY COURSE PLAN

Week	Subject
1	<i>Introduction to Database Management Systems; Concepts and Architecture</i>
2	<i>Data Modeling Using the Entity-Relationship (ER) Modeling</i>
3	<i>Data Modeling Using the Entity-Relationship (ER) Modeling</i>
4	<i>Enhanced Entity-Relationship (EER) Modeling</i>
5	<i>Enhanced Entity-Relationship (EER) Modeling</i>
6	<i>Relational Data Model and Relational Database Constraints</i>
7	<i>ER and EER to Relational Mapping; Examples and Homeworks</i>
8	Internal Exam
9	<i>Functional Dependencies and Normalisation Forms for Relational Databases</i>

WEEKLY COURSE PLAN

Week	Subject
10	<i>Examples of Normalisation Forms</i>
11	<i>Query Representation; Query Types; Relational Algebra and Calculus</i>
12	<i>Query Representation; Query Types; Relational Algebra and Calculus; Examples</i>
13	<i>Disk Storage; Basic File Structures and Hashing; Indexing Structures for Files</i>
14	<i>Make-Up Exam</i>

EVALUATION

	Number	Effective Proportion %
Midterm Exams	1	50
Quiz		
Homework	1	10
Term Projects		
Term Papers		
Laboratory		
Other		
Final Exam	1	40

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Questions and Answers

Thank you

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